# Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

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## 1. Introduction

For centuries, metals have been among the materials on which human societies are built, from cities and transport networks to power grids, machinery and the technologies of modern economies. Their uses have widened with each wave of industrial development, and the sector remains large: according to McKinsey & Company's Global Materials Perspective, the metals and mining industry generated about USD 3.0 trillion in revenue in 2024 after a decline of 6 percent and grew by only about 1 percent in 2025 (McKinsey & Company, 2025, 2026).

Within this broad market, high performance alloys serve the conditions that ordinary materials cannot withstand, such as the hot section of a jet engine, a sour gas well, a chemical reactor or a permanent implant. These engineered materials combine nickel, cobalt, chromium, molybdenum, titanium, tungsten and other elements to provide strength at high temperature, corrosion and oxidation resistance or biocompatibility, and they reach industry as qualified bar, billet, sheet, strip, wire, tube and powder. Their raw materials, however, come from a small number of countries, as Indonesia mines about 63 percent of the world's nickel, the Democratic Republic of the Congo about 73 percent of its cobalt and China most of its tungsten (International Energy Agency, 2026b; U.S. Geological Survey, 2026a, 2026e).

The outlook for these alloys to 2035 is positive but uneven, and its size is still poorly measured. Commercial estimates place the segment between about USD 10 billion and USD 14 billion in 2025, a spread that reflects different market boundaries rather than different markets (IMARC Group, 2025; The Insight Partners, 2025). Part of the future demand is already visible in engine and turbine programmes, energy investment and the supply of alloying metals that export policy increasingly shapes, while another part depends on technologies such as hydrogen turbines, advanced reactors and additive manufacturing that official statistics do not yet capture. This study therefore measures the first part from official production, trade and industry data and treats the second as a set of emerging opportunities whose evidence and timing are assessed openly rather than hidden in a single headline figure.

## 2. Report Objective and Scope

This project studies selected metals that are essential to high performance alloys and follows them from the mine to the industrial applications that depend on them, in order to assess how their markets may develop through 2035. It asks one central question: how do the concentration of metal supply, the conversion of metals into qualified product forms and the needs of seven industries determine where the most attractive opportunities in high performance materials will lie through 2035? Four subsidiary questions structure the answer, and each maps onto one part of the report. The first asks where the ten metals are mined and refined, how exposed that supply is to concentration and export policy, and in which qualified

forms the metals reach industry. The second asks which industries and regions drive demand for these forms today and how large the measurable market is. The third asks how large demand could become by 2035, how certain that estimate is and which emerging applications could add to it, and the fourth asks where supply security, growth and entry barriers combine into the most attractive opportunities.

The scope covers ten metals, namely nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese, and seven industries, namely automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The analysis is global, with regional comparisons for Europe, North America and Asia, and it uses the most recent official data available in November 2026. Mine output, refined metal, alloy melt, mill products and industry revenue measure different stages of the value chain, so the report never adds them into one aggregate market value. The outlook has two parts, a quantified estimate of demand for established applications and a structured assessment of emerging applications that may matter by 2035 but are not yet visible in official statistics.

The report argues that value in this market is created by converting metals into qualified products rather than by mining them. Nickel is in surplus, while cobalt, tungsten and molybdenum are constrained by export policy, and the countries that convert metals into qualified bar, strip and tube earn trade surpluses that the mining countries do not. Chapter 3 explains how this argument is tested, chapters 4 to 7 follow the metals from supply through industry demand and trade to growth, chapter 8 identifies where the most attractive opportunities lie and chapter 9 concludes.

## 3. Introduction to the Analytical Framework and Methodology

The analysis follows a metal value chain that runs from material through product form, application, industry and geography to market size and growth, and Table 1 shows how the chapters cover its links. A figure is only compared with a figure from the same stage, so contained metal is kept apart from gross weight and output from capacity. The evidence starts with structured statistics from the USGS, UN Comtrade, Eurostat PRODCOM, the World Bank, OECD, UNIDO and the IEA, continues with industry associations, official outlooks and the filings of 61 companies, and uses commercial market reports only as a cross check. Every key figure is confirmed by a second independent source, and the research was organised in specialised workstreams for each metal, each industry and the company universe, whose findings were consolidated into a shared workbook in which every value carries its unit, stage, data type, sources and formula.

Table 1: The metal value chain and its evidence

| Link | Guiding question | Main evidence | Chapter |
|---|---|---|---|
| Material | Where is the metal mined and refined, and how concentrated is supply? | USGS, IEA critical minerals data, World Bank prices | 4 |
| Product form | In which qualified forms does the metal reach industry? | Eurostat PRODCOM, producer filings | 4 |
| Application | Which components need the material, and why? | Standards, associations, filings | 4 and 5 |
| Industry and geography | Which industries and regions buy, and how fast do they grow? | Airbus, Boeing, IEA, OICA, Cefic, SEMI, OECD | 5 |
| Market size | How large is measurable demand today? | Apparent consumption, trade values, end use statistics | 5 and 6 |
| Growth | How large could demand be in 2035, and what could add to it? | Calibrated demand model and emerging opportunities register | 7 |

*Source: team framework.*

The outlook to 2035 is built in two tiers that are never mixed. The first tier is a demand model for established applications, which multiplies the activity of each industry, such as engine deliveries, gigawatts of gas turbines or joint replacements, by the metal content per unit and by the share of units that use a high performance alloy. This model is calibrated to reproduce published history, back tested by fitting it on data up to 2019 and comparing its projection for 2020 to 2025 with actual outcomes, and corrected for the past optimism of official outlooks, while a Monte Carlo simulation yields the 10th, 50th and 90th percentiles of 2035 demand and a supply overlay tests them against export quota and licence shocks.

The second tier covers opportunities that the data cannot yet confirm, such as hydrogen fuelled turbines, supercritical carbon dioxide recuperators, molten salt and small modular reactors, fusion magnet substrate tape or additively manufactured engine parts. Each candidate is drawn from patents, pilot plants, research programmes, standards under development or announced projects and assessed on its technology readiness, the strength of its evidence, the alloys and product forms it would need and the date by which it could reach commercial volume. These options are reported as a separate register with qualitative ranges and signposts, such as a final investment decision or a first serial order, that would move them into the model, and chapter 8 scores both tiers on the same eight criteria while flagging the evidence behind each score, so that measured positions are never confused with options that should be monitored rather than funded at scale.

## 4. Metal Markets: Materials, Product Forms and Applications

Every high performance alloy begins as raw materials that the alloy maker does not control, so the analysis starts with the metals. This chapter follows ten metals from material through product form to application, the first three links of the framework, and Table 2 condenses them into a scorecard that already shows the pattern: supply is concentrated in a few countries, while the qualified product forms are made elsewhere.

Table 2: Metal scorecard along the first three links, 2025

| Metal | Material: world output and largest producer | Price signal | Main product forms | Key high performance applications | Supply risk to 2035 |
|---|---|---|---|---|---|
| Nickel | 3.90 Mt contained (IEA 4.21 Mt), Indonesia about 67% | USD 15,162/t, 41% below the 2022 average | Strip, sheet, plate, bar, wire, tube | Corrosion and heat resistant alloys, superalloys | Medium, Indonesian ore quotas |
| Cobalt | 310 kt contained, DR Congo 73% | Metal up 130% in 2025 | Alloy grade metal, bar, sheet, powder | Superalloys, CoCr implants, wear alloys | High, Congolese export quota |
| Chromium | 51 Mt chromite ore, South Africa about 45% | Chromium metal up, ferrochrome down | Ferrochrome, chromium metal | Every corrosion and oxidation resistant alloy | Medium, smelting in China |
| Molybdenum | 260 kt contained, China about 37% | USD 51/kg oxide, up 7% | Oxide, ferromolybdenum, metal | Pitting resistance in C-276, 625 and 6Mo grades | Medium, Chinese licences on powders |
| Titanium | 370 kt sponge, China about 70% | USD 12/kg U.S. sponge import value | Sponge, ingot, bar, plate | Airframes, engine cold sections, implants | High for aerospace grade sponge |
| Tungsten | 85 kt contained, China about 79% | Rising under export licences | APT, carbide, metal | Cast superalloys, heat resistant sheet | High, Chinese quotas and licences |
| Zirconium | 1.2 Mt concentrates, Australia about 33% | USD 1,800/t premium zircon | Sponge, plate, tube | Acid service equipment, fuel cladding | Medium, nuclear grade is captive |
| Copper | 23.2 Mt mine, 28.6 Mt refined | USD 9,947/t, LME record in 2026 | Cathode, wire rod, strip, foil | Busbars, connectors, CuNi alloys | Medium, mine supply falls by 2035 (IEA) |
| Aluminium | 74 Mt primary, China about 61% | USD 2,632/t | Sheet, plate, extrusions | Competing material in heat exchangers, busbars | Low |
| Manganese | 20 Mt contained, South Africa about 38% | USD 4.5 per mtu ore | Ferromanganese, metal | Alloying addition in stainless | Low |

*Sources: USGS Mineral Commodity Summaries 2026, IEA, World Bank Pink Sheet, Cobalt Institute, ICSG, IAI and IMOA (Cobalt Institute, 2026; International Copper Study Group, 2026a, 2026b; International Energy Agency, 2026b; Ministry of Commerce of the People's*

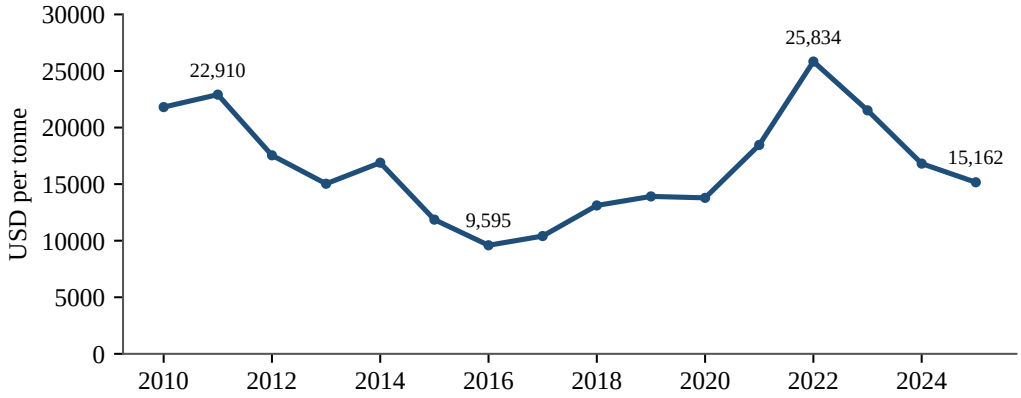
*Republic of China, 2025a; The Oregon Group, 2026; U.S. Geological Survey, 2026a, 2026c, 2026d, 2026e, 2026f, 2026g, 2026i, 2026l; World Bank, 2026a, 2026b, 2026c). Shares are team calculations on rounded USGS totals, except the Congo share, which USGS publishes.*

The scorecard sorts the ten metals into three groups that structure the rest of the report. Nickel and chromium are the backbone of every alloy and are available in volume, cobalt, tungsten, molybdenum and aerospace grade titanium are scarce because of policy or qualification, and copper, aluminium and manganese matter mainly as complementary or competing materials, so nickel, the base of the whole alloy family, comes first.

## 4.1 Nickel

Nickel is the foundation of the high performance alloy industry, and within a decade its supply has moved to a single country. USGS estimates world mine output at 3.90 Mt of contained nickel in 2025, with 2.60 Mt or about two thirds from Indonesia, while the IEA puts the total at 4.21 Mt and Indonesia at 63 percent of mining and 45 percent of refining, so the series differ in level but not in concentration (International Energy Agency, 2026b; U.S. Geological Survey, 2026h). Primary production reached 3.875 Mt against usage of 3.577 Mt, yet just over half of the primary supply in China and Indonesia was nickel pig iron for stainless steel rather than the class 1 metal that vacuum melted alloys require (International Nickel Study Group, 2025, 2026). Alloy makers therefore depend on a few class 1 sources such as Vale in Canada, Glencore Nikkelverk in Norway and Nornickel in Russia, and Russian metal produced on or after 13 April 2024 may no longer enter the United States or be warranted on metal exchanges under US and UK sanctions (U.S. Department of the Treasury, 2024; United Nations Statistics Division, 2026a).

Figure 1: The nickel price has fallen by about 41 percent from its 2022 annual average




*Source: World Bank Pink Sheet, annual averages of monthly prices, team calculation (World Bank, 2026a, 2026b).*

This oversupply explains the price, which averaged USD 25,834 per tonne in 2022 and USD 15,162 in 2025 in the World Bank series, against a USGS estimate of USD 15,000 (U.S. Geological Survey, 2026h; World Bank, 2026a). Indonesia now manages output through its annual ore quota, cutting the 2026 quota to 260 to 270 Mt of ore before raising it to 290 to 300 Mt, and the International Nickel Study Group cut its 2026 surplus forecast to 19 kt after 299 kt in 2025, so the price of the raw material is now set in Jakarta as much as in London (Argus Media, 2026a; Indonesia Business Post, 2026; International Nickel Study Group, 2026).

Nickel reaches industry in two fundamentally different product forms, and the gap between them reveals where value is created. Stainless steel absorbs most nickel, with 64.2 Mt of gross melt shop output in 2025, and the IEA expects nickel demand outside energy technologies, which covers stainless steel and alloys, to grow only from 2,906 kt in 2025 to 3,255 kt in 2035 (International Energy Agency, 2026b; worldstainless, 2026). Nickel alloys are sold as strip, sheet, plate, bar, wire and tube, and German exports of nickel plate, sheet and strip reached 17,775 t worth USD 578 million in 2024, about USD 32,500 per tonne or twice the price of the contained metal, a tonnage that Eurostat confirms within 2 percent (Eurostat, 2026a; United Nations Statistics Division, 2026c; World Bank, 2026a). Buyers pay for qualification and form rather than for nickel, a pattern that cobalt makes sharper still.

## 4.2 Cobalt

Cobalt is the scarcest of the major superalloy elements, and its scarcity is created by policy rather than by geology. USGS estimates world mine output at 310 kt of contained cobalt in 2025, close to the IEA figure of 320 kt, and the Democratic Republic of the Congo produced about 230 kt or 73 percent, while China refined 78.6 percent of the world's 240 kt of refined cobalt (Cobalt Institute, 2026; International Energy Agency, 2026b; U.S. Geological Survey, 2026e). In February 2025 Congo banned exports and then imposed a quota of up to 96,600 t a year for 2026 and 2027, CMOC, which mined 117,549 t in 2025, received an export entitlement of only 31,200 t, and the European cobalt metal price rose 130 percent during 2025 (CMOC Group, 2026; Cobalt Institute, 2026; Mining.com, 2026; U.S. Geological Survey, 2026e). The IEA expects mine supply to fall to 225 kt by 2035 against demand of 300 to 322 kt, so the shortage is structural (International Energy Agency, 2026b).

The alloy industry is a small customer in this market, but it buys the most demanding product form. Batteries for vehicles and portable electronics took 206 kt of cobalt in 2025 against only 20 kt for superalloys, which buy mostly high purity metal from a few Western refineries (Cobalt Institute, 2026). Cobalt reaches its applications as superalloy remelt stock for turbine blades and combustors, as cobalt chromium bar and powder for hip and knee implants, and as wear resistant alloys. Because these applications cannot easily switch to another metal, a quota decided in Kinshasa becomes a cost and continuity risk in Toulouse and Cincinnati, and chromium shows a similar dependence one step later in the chain.

## 4.3 Chromium

Chromium gives every corrosion and heat resistant alloy its protective oxide layer, yet its value chain is split between the countries that mine it and the country that smelts it. World chromite ore output reached an estimated 51 Mt gross weight in 2025, of which South Africa mined about 23 Mt on USGS figures and 25.2 Mt on the differently defined dashboard of the International Chromium Development Association (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026d). The same association reports 9.9 Mt of ferrochrome alloy for China, a smelter product that cannot be set against ore tonnes, from a country with almost no chromite (International Chromium Development Association, n.d.). Stainless steel absorbs most of this ferrochrome, and the United States imports 79 percent of its chromium needs (U.S. Geological Survey, 2026d; worldstainless, 2026).

The product form that high performance alloys require is far more specialised than the ferrochrome that stainless mills consume. Vacuum melted nickel alloys use low carbon ferrochrome and aluminothermic chromium metal, a small market whose U.S. price rose to an estimated USD 5.90 per pound in 2025 while the ferrochrome price fell (U.S. Geological Survey, 2026d). A single qualified source of this small input can therefore halt a vacuum melt shop, a vulnerability that molybdenum repeats in an even more value intensive form.

## 4.4 Molybdenum

Molybdenum is what turns ordinary stainless steel into the alloys that survive hot acids and seawater, and its intensity varies enormously between grades. One tonne of 316L stainless contains about 21 kg of molybdenum, whereas one tonne of Alloy C-276 contains about 160 kg and Alloy 625 about 90 kg, so a small volume of nickel alloy carries a large molybdenum bill (ATI, n.d.; Haynes International, 2023; Outokumpu, 2025).

Supply is tied to copper mining, which limits its response to price, and USGS estimates world mine output at 260 kt of contained molybdenum in 2025, while the International Molybdenum Association reports 672.6 million pounds, about 305 kt in a team conversion, and almost the entire gap sits in China (International Molybdenum Association, 2026; U.S. Geological Survey, 2026g). According to USGS only China and the United States produce from primary molybdenum mines, so most other supply follows the copper mine plan, and since February 2025 China has licensed exports of molybdenum powders rather than of the oxide and ferromolybdenum that melt shops buy (Ministry of Commerce of the People's Republic of China, 2025a; U.S. Geological Survey, 2026g). The U.S. oxide price rose 7 percent to an estimated USD 51 per kg in 2025, while European use fell 7 percent with the weak chemical cycle (International Molybdenum Association, 2026; U.S. Geological Survey, 2026g). Molybdenum thus links corrosion alloys to copper mining and Chinese policy, whereas titanium competes with nickel alloys more than it feeds them.

## 4.5 Titanium

Titanium replaces nickel alloys in lighter and cooler parts while sharing their aerospace buyers and qualification bottlenecks, and its mineral and metal markets barely overlap, since world output of ilmenite and rutile reached 9.8 Mt of TiO2 in 2025, mostly for pigment, while sponge metal reached only 370 kt (U.S. Geological Survey, 2026i, 2026j). China produced about 260 kt of that sponge and holds 320 kt of the 470 kt of world capacity, but Chinese sponge is not the qualified source for Western engines, and the United States has no operating sponge capacity (U.S. Geological Survey, 2026i).

The scarce product form is aircraft grade sponge and the ingot, bar and plate made from it, as Osaka Titanium is raising aircraft grade capacity from 40 kt to 50 kt by fiscal 2027, ATI quotes lead times of about 20 months for premium titanium, and Europe melts and fabricates titanium without producing sponge, selling 22.6 kt of wrought titanium in 2024 against imports of 31.7 kt (ATI, 2026b; Eurostat, 2024; Osaka Titanium Technologies, 2025). The same shortage of qualified capacity constrains nickel superalloys, which strengthens every qualified melter, and tungsten shows how quickly one government can reshape such a market.

## 4.6 Tungsten

Tungsten is the clearest example of a metal whose supply one state controls. World mine output reached an estimated 85 kt of contained tungsten in 2025, of which China produced about 67 kt, or 79 percent (U.S. Geological Survey, 2026k). China sets national mining quotas, with a first batch of 58,000 t of concentrate for 2025, and since February 2025 it has required export licences for ammonium paratungstate, tungsten oxides, unsintered carbide and specified tungsten alloys, while sintered carbide tools and tungsten wire remain outside the control (Ministry of Commerce of the People's Republic of China, 2025a, 2025b; Ministry of Natural Resources, 2025).

Most tungsten ends up as carbide for cutting and mining tools, which takes 65 percent of first use, while steels and superalloys take 14 percent (International Tungsten Industry Association, n.d.). In superalloys tungsten strengthens cast turbine alloys and heat resistant sheet, so a licence decision in Beijing soon reaches the gas turbine hot section, and Sandvik's powder business more than doubled orders and revenues on higher tungsten prices in the third quarter of 2026 (Sandvik, 2026). Tungsten thus closes the group of policy constrained metals before the niche and competing metals of section 4.7.

## 4.7 Other Selected Metals

The remaining metals complete the picture in two ways, because zirconium and the minor superalloy elements are niche inputs, while copper, aluminium and manganese are large markets with a small high performance corner.

### 4.7.1 Zirconium

Zirconium shows why mineral output and specialty metal must be kept apart, since world production of zirconium concentrates reached about 1.2 Mt gross weight in 2025, led by Australia with 400 kt, mostly for ceramics, foundry sand and refractories (U.S. Geological Survey, 2026l). The metal market is far smaller and divides into nuclear fuel cladding, which fuel makers and integrated producers such as ATI control themselves, and zirconium plate and tube for acid service, which is the open part of the market (ATI, 2026a).

### 4.7.2 Copper

Copper links the alloy market to electrification, although high performance products are a small part of it. World mine production reached about 23.2 Mt in 2025 on ICSG and IEA figures alike, flat rolled products take only 11 percent of first use against 63 percent for wire, and the price set LME records above USD 14,500 per tonne in 2026 (International Copper Study Group, 2025, 2026a; International Energy Agency, 2026b; The Oregon Group, 2026). The IEA expects mine supply from current projects to fall to 20.7 Mt by 2035 while demand rises from 27.8 Mt to 32.4 Mt under current policies, and it sizes the primary deficit in 2035 at about 25 percent (International Energy Agency, 2026a, 2026b). Data centres illustrate the new demand, since a 1 MW rack at 54 VDC needs up to 200 kg of copper busbar, and the specialised forms are copper nickel and high conductivity strip (NVIDIA, 2025a).

### 4.7.3 Aluminium

Aluminium matters here mainly as a competitor, because it replaces heavier alloys wherever temperature and corrosion allow it. World primary production reached about 74 Mt in 2025, of which China produced 61 percent, while European primary output has shrunk to about 955 kt against demand of about 9 Mt a year (European Aluminium, n.d.; Federation of Aluminium Consumers in Europe, 2024; International Aluminium Institute, 2026; U.S. Geological Survey, 2026c).

### 4.7.4 Manganese

Manganese is an alloying addition whose tonnage follows carbon steel, and world mine output reached about 20 Mt of contained manganese in 2025, led by South Africa, Gabon, Ghana and Australia, which together supplied about 81 percent, and the ore price fell to about USD 4.5 per metric ton unit, so it is neither a supply risk nor a growth market for alloy producers (International Manganese Institute, 2026; U.S. Geological Survey, 2026f).

#### 4.7.5 Additional Metals

Iron matters through FeNi alloys such as FeNi36 for display masks and FeNi42 for lead frames, whose value is set by rolling capability rather than by iron supply. Niobium and rhenium matter for superalloys, because Alloy 718 contains about 5 percent niobium and single crystal turbine blades contain rhenium, of which only about 81 t were mined worldwide in 2025, mostly as a byproduct of copper and molybdenum (RTX, 2026; U.S. Geological Survey, 2026b). The analysis now turns to the industries that convert these materials into demand for alloys.

## 5. Industry Needs: Demand, Geography and Market Size

Metals create value only when an industry needs them, so the chain now moves from material to demand. Each of the seven industries follows the same links, from size and regions to the applications and product forms through which high performance metals enter it. Because industry revenue is not alloy demand, each section uses the most direct measurable indicator and closes its first subsection with the best measured size at a named stage of the chain. Where no alloy tonnage is published, the industry enters chapter 7 only as a team index, and appendix A5 holds the indicator tables. To avoid double counting, electrolysers and fuel cells belong to energy, process vessels to chemicals, and sensors and magnetic parts to electronics.

### 5.1 Automotive

Automotive is the clearest case of high performance alloys losing ground, because electrification removes the turbochargers, exhaust sensors, glow plugs and gaskets that have long consumed heat resistant nickel alloys.

#### 5.1.1 Demand, Geography and Market Size

Vehicle output still grows but shifts toward China and electric cars, as the world built 96.4 million motor vehicles in 2025, 34.5 million of them in China, while EU car registrations of 10.8 million stayed 17 percent below their 2019 level (European Automobile Manufacturers' Association, 2026a; International Organization of Motor Vehicle Manufacturers, 2026). Electric cars exceeded 20 million sales in 2025, a quarter of the market, and the IEA expects about half of global car sales and more than 90 percent in China and the EU to be electric by 2035 (International Energy Agency, 2026c). The battery offers little compensation, since lithium iron phosphate cathodes, which contain no nickel, took over 55 percent of electric vehicle batteries in 2025 (International Energy Agency, 2026c).

European policy may soften the shift, since in October 2026 the European Parliament's industry and transport committees backed a 95 percent cut in new car CO2 emissions for 2035, or 87.5 percent with credits for European low carbon steel, ahead of a plenary vote expected in

December 2026 (electrive, 2026). The best measured metal figure is therefore a battery figure, IEA nickel demand from electric vehicles of 439 kt in 2025 and 1,294 kt to 1,536 kt in 2035, which is cathode material, while engine alloy tonnage is unmeasured and declining (International Energy Agency, 2026b).

### 5.1.2 Applications and Opportunities

Engine alloys decline slowly, because hybrids and gasoline engines keep turbochargers in production for another decade. Turbocharger output reached close to 50 million units in 2025, and Garrett expects it to fall back toward the 2022 level by 2030 while gasoline turbo penetration peaks at 54 percent in 2032 (Garrett Motion, 2026). BorgWarner still won three gasoline turbocharger programmes in October 2026 with production from 2027 (BorgWarner, 2026).

Table 3: High performance metal applications in automotive, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Current sensors and magnetic shielding in electric drives and chargers | Soft magnetic FeNi strip |
| Stable to 2030 | Turbocharger parts for hybrids and gasoline engines | Heat resistant nickel alloy bar and strip |
| Falls | Glow plugs, diesel exhaust parts, valves and injectors | Nickel alloys, heat resistant stainless strip |

*Sources: team assessment (BorgWarner, 2026; Garrett Motion, 2026; International Energy Agency, 2026c).*

Implications for suppliers:

- Treat engine alloys for cars as a cash business to 2030 and avoid new capacity for diesel only parts.
- Shift product development toward soft magnetic FeNi strip for sensors and shielding in electric drives.

## 5.2 Oil and Gas

Oil and gas is a stable market for high performance metals, in which investment moves from oil to gas and sour gas, deep water and LNG need alloys that ordinary steel cannot replace.

### 5.2.1 Demand, Geography and Market Size

Upstream investment holds its level, but much of it only defends existing output. World upstream oil and gas investment fell from USD 625 billion in 2019 to USD 542 billion in 2025

and about USD 546 billion in 2026, and about 40 percent of upstream spending only slows the decline of existing fields (International Energy Agency, 2025b, 2026h, 2026j). Behind this flat total, North America remains the largest region at USD 165 billion in 2026 but has fallen from USD 219 billion in 2019, while the Middle East is the largest region spending above its 2019 level, at USD 93 billion against USD 75 billion, and its gas investment roughly doubled from USD 22.9 billion to USD 45.4 billion (International Energy Agency, 2026j). Central and South America grew fastest, from USD 42 billion to USD 68 billion, and Sub Saharan Africa recovers by 12 percent in 2026, while LNG trade could grow from 560 bcm in 2024 to 880 bcm in 2035 under current policies (International Energy Agency, 2025c, 2026h, 2026j). Upstream investment is thus the best measured stage, since no public source counts the tonnes of nickel alloy, duplex or clad pipe that the industry buys.

### 5.2.2 Applications and Opportunities

The real barrier to entry in this market is customer approval rather than metallurgy. ISO 15156 defines which corrosion resistant alloys in which metallurgical condition suit sour service, so a listed grade is a commodity, whereas each producer and product form must pass NORSOK M-650 qualification and operator approval (International Organization for Standardization, 2020; Standards Norway, 2011). Orders then flow through the approved vendor lists of EPC contractors and subsea majors such as SLB and TechnipFMC, which took USD 10.1 billion of subsea orders in 2025 (SLB, 2026; TechnipFMC, 2026). Product forms reach from nickel alloy bar for wellheads and completion tools to umbilical tube, where Alleima offers hyper duplex SAF 3007 as successor to the standard super duplex SAF 2507, and clad pipe, for which BUTTING reports turnover of EUR 880 million (Alleima, n.d.; BUTTING, 2026).

Table 4: High performance metal applications in oil and gas, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Clad and lined pipe, subsea equipment, LNG cryogenic parts | Nickel alloy strip, plate, welding wire, FeNi36 |
| Stable | Well tools, wellheads, valves and fittings for sour gas | Nickel alloy bar, billet, plate |

*Sources: team assessment (Alleima, n.d.; BUTTING, 2026; International Energy Agency, 2026h; TechnipFMC, 2026).*

Competitors invest despite the flat market, as Outokumpu adds about 13,000 t of high nickel flat capacity at Avesta from 2028, which shows how much approved positions are worth (Outokumpu, 2026b).

Implications for suppliers:

- Win places on the vendor lists of Middle East and South American operators and of their EPC contractors and subsea majors.
- Grow strip and welding wire for clad pipe makers once NORSOK M-650 qualification is secured.

## 5.3 Energy and Environmental Technology

Energy technology holds both the most urgent shortage and the most visible disappointment in the market, because gas turbines cannot be built fast enough while hydrogen has so far produced more announcements than orders.

### 5.3.1 Demand, Geography and Market Size

Gas turbines are the tightest pocket, because data centres and grid stability created more orders than the industry can build. Siemens Energy is sold out until fiscal 2028 and will raise its gas turbine capacity from about 17 GW in fiscal 2024 to above 30 GW a year (Siemens Energy, 2025). Investment in gas fired power almost doubled from USD 64 billion in 2019 to USD 112 billion in 2025, and data centres could use about 945 TWh of electricity in 2030, more than twice their 2024 level (International Energy Agency, 2025a, 2026j).

Hydrogen shows the opposite picture, as low emissions production was only about 1 Mt of more than 100 Mt of demand in 2025 and the announced 2030 pipeline shrank to 27 Mt (International Energy Agency, 2026d, 2026e). The IEA project database counts 4.3 GW of electrolysis in operation, 19.3 GW at final investment decision or under construction and 858 GW at concept or feasibility stage, and alkaline electrolysers, the nickel route, make up 60 percent of the capacity being built (International Energy Agency, 2026g). thyssenkrupp nucera, the leading European alkaline electrolyser maker, saw green hydrogen orders fall from EUR 356 million to EUR 26 million in fiscal 2024/25 (thyssenkrupp nucera, 2025).

Nuclear power offers a slower but steadier pull, with 81 reactors under construction, 40 of them in China, and a United States framework of October 2025 covering at least USD 80 billion of AP1000 reactors (Westinghouse Electric, 2025; World Nuclear Association, 2026b). At the metal stage, the IEA puts nickel demand for hydrogen technologies at only 2.3 kt in 2025 and 5.2 kt in 2035 in its stated policies case, or 32 kt in its high demand case, while gas turbine superalloy tonnage is not published (International Energy Agency, 2026b).

### 5.3.2 Applications and Opportunities

The three resulting opportunities differ in timing more than in size, as Table 5 shows. Gas turbine parts are needed now, whereas fuel cells remain a weak metal market, because Bloom Energy, which doubles its solid oxide output to 2 GW a year by the end of 2026, does not disclose its interconnect material (Bloom Energy, 2026). In nuclear power Alleima has

reopened a Sandviken tube mill for steam generator tube, which places alloy producers as suppliers to tube makers rather than in fuel cladding (Alleima, 2026a).

Table 5: High performance metal applications in energy technology, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows strongly | Gas turbine blades, vanes, combustors and repair | Superalloy remelt stock, combustor sheet, welding wire |
| Grows from 2030 | Nuclear steam generator tube | Alloy 690 bar and hollows for tube makers |
| Option | Solid oxide cell interconnects, alkaline electrolyser electrodes | Coated ferritic strip, nickel strip and mesh |

*Sources: team assessment (International Energy Agency, 2026g; Siemens Energy, 2025; thyssenkrupp nucera, 2025; World Nuclear Association, 2026b).*

Implications for suppliers:

- Qualify superalloy remelt stock and combustor sheet with turbine casting houses while capacity is sold out to 2028.
- Treat electrolysers, fuel cells and nuclear tube as options that are released only against signed customer volumes.

## 5.4 Aerospace

Aerospace is the growth engine of high performance metals, because every jet engine hot section is made of nickel and cobalt superalloys that the supply chain struggles to melt and forge fast enough.

### 5.4.1 Demand, Geography and Market Size

Fleet growth sets the long term path, as Airbus expects the passenger fleet to grow from 23,310 aircraft in 2025 to 45,550 in 2045, which requires 42,060 new aircraft, while Boeing expects 43,625 deliveries on a wider definition (Airbus, 2026a; Boeing, 2026). Engines grow faster than aircraft, as CFM delivered 1,802 LEAP engines in 2025, up 28 percent (GE Aerospace, 2026a; Safran, 2026a). The aftermarket adds profit, since aviation services grow from USD 209 billion in 2026 to USD 422 billion in 2045 and materials drive over 85 percent of off wing maintenance cost (Airbus, 2026b).

Defence adds a third stream, with military expenditure of USD 2,887 billion in 2025 and GE delivering 635 defence engines against 490 a year earlier (GE Aerospace, 2026a; Stockholm International Peace Research Institute, 2026). Although deliveries shift toward Asia, qualified melting, forging and casting capacity stays in the United States, the United Kingdom, France

and Germany. The best measured size at the metal stage is cobalt, as the Cobalt Institute counts 11,499 t of cobalt in aviation in 2025 within 20 kt used in superalloys, and the U.S. Geological Survey reports that 51 percent of United States cobalt consumption goes into superalloys, mainly for aircraft engines, while nickel alloy tonnage is not published (Cobalt Institute, 2026; U.S. Geological Survey, 2026e).

### 5.4.2 Applications and Opportunities

Rotating parts set the highest barrier among the applications in Table 6. Disc and shaft billet is premium melted by vacuum induction melting followed by electroslag and vacuum arc remelting, as at ATI in Lockport and Special Metals in Huntington, and the most demanding discs use superalloy powder, whose quality risk the powder metal matter on Pratt and Whitney's geared turbofan exposed (ATI, 2026a; RTX, 2026; Special Metals Corporation, n.d.). Each approved melt route is frozen in the engine maker's own specification on top of AMS standards and NADCAP processes, and GE Aerospace states that qualifying a new supplier takes substantial time, so a first rotating approval takes several years (GE Aerospace, 2026a).

Table 6: High performance metal applications in aerospace, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows strongly | Discs, shafts, casings and rings for LEAP, GE9X, Trent and military engines | Triple melt superalloy bar and billet, powder for discs, sheet |
| Grows | Engine repair and overhaul | Welding wire, repair alloys |
| Grows | Additively manufactured combustor and nozzle parts | Superalloy powder |
| Stable | Combustor and wear parts where nickel alloys replace cobalt | Cobalt alloy sheet and plate |

*Sources: team assessment (Airbus, 2026a; ATI, 2026a; Safran, 2026a).*

The shortage shows in the market leaders, who lock in supply and build captive capacity. ATI earned USD 1,762.9 million from commercial jet engines in 2025 under long term agreements with GE Aerospace, Safran, Rolls-Royce and Pratt and Whitney and quotes forging lead times of more than two years (ATI, 2026a, 2026b). Safran committed EUR 150 million to a 30,000 t press at Gennevilliers for about 14,000 hot section parts a year from 2029 (Safran, 2026b). The window for a new qualified European source is open today but narrows as these investments come on stream around 2030.

Implications for suppliers:

- Start part level qualifications with engine makers and independent forgers now, because rotating approvals take several years.
- Build the repair wire and powder business alongside, because the maintenance market grows by 3.8 percent a year (Airbus, 2026b).

## 5.5 Chemical Processing

Chemical processing is the traditional heartland of corrosion resistant alloys, but in Europe it has become a maintenance market while new plants are built in Asia.

### 5.5.1 Demand, Geography and Market Size

Europe's chemical industry is shrinking in volume and running with spare capacity. EU27 chemical sales were about EUR 635 billion in 2024, but Europe's share of world chemical sales fell from 21 percent in 2009 to 13 percent, while China's rose from 24 to 46 percent (European Chemical Industry Council, 2025a, 2025b). The EU chemical production index fell 3.2 percent in 2025, and capacity utilisation was 75.0 percent in the third quarter of 2026 against a long term average of 81.3 percent (European Chemical Industry Council, 2026b; Eurostat, 2026c). New capital follows growth to Asia, as BASF expects over 80 percent of chemical industry growth to be in Asia Pacific by 2035 and opened its Zhanjiang Verbund site in November 2025 (BASF, 2026). The closest measured size at the metal stage is EU apparent consumption of nickel plate, sheet, strip and foil, 57.8 kt in 2024 across all end uses, of which chemical processing holds an unmeasured share (Eurostat, 2025d).

### 5.5.2 Applications and Opportunities

Corrosion resistant alloys remain indispensable wherever stainless steel fails, from sulphuric and hydrochloric acid to chlor alkali service. Table 7 shows that European demand is mostly replacement, while new construction and its plate and clad plate move to Asia.

Table 7: High performance metal applications in chemical processing, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Stable | Replacement heat exchangers, vessels and piping in European plants | C-276, C-22 and 59 type NiCrMo alloys and Alloy 625 as plate, sheet and welding consumables |
| Grows | New plants in China, India and Southeast Asia | Plate, clad plate, zirconium plate and tube |
| Selective | Battery material plants, where titanium and linings dominate | Plate for selected duties, welding consumables |

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Falls | New build in European base chemicals | Plate |

*Sources: team assessment (Alleima, 2026b; BASF, 2026; European Chemical Industry Council, 2026b).*

The leading suppliers are moving closer to that new demand, and Alleima is investing SEK 220 million in finishing at Mehsana in India to serve Asian chemical and petrochemical customers (Alleima, 2026b).

Implications for suppliers:

- Defend European replacement demand with plate and welding consumables sold on documented life cycle cost.
- Follow new construction to India and Southeast Asia through the engineering contractors that build Asian plants.

## 5.6 Electronics and Electrical Engineering

Electronics has the largest headline numbers and the smallest metal volumes, because booming chip and data centre markets use precision strip, foil and high purity alloys measured in tonnes.

### 5.6.1 Demand, Geography and Market Size

The semiconductor boom is real but must be read carefully before it is translated into metal. Semiconductor sales reached USD 795.6 billion in 2025, up 26.2 percent, and WSTS forecasts USD 1.51 trillion for 2026 mainly because memory sales more than triple, so sales value rises without a comparable rise in metal volume (World Semiconductor Trade Statistics, 2026). Equipment spending is the closer indicator, and SEMI expects 300 mm fab equipment spending of USD 149 billion in 2026 and USD 227 billion in 2029, with fabs in Taiwan, Korea, China and increasingly the United States (SEMI, 2026a, 2026c). Germany's electrical and electronics industry added EUR 112.2 billion of value in 2023 but showed flat real output in 2025 (OECD, 2026; ZVEI, 2026). No public source measures FeNi or high purity alloy tonnage for electronics, whose FeNi growth comes from electronics and electric drive sensors rather than from engine alloys.

### 5.6.2 Applications and Opportunities

In data centres, NVIDIA states that a 1 MW rack at 54 VDC needs up to 200 kg of copper busbar and plans 800 VDC distribution with its Kyber racks in 2027, with partners such as

Infineon, Vertiv, Eaton and Schneider Electric, and Vertiv raised its copper hedge to 12,946 t by September 2026 (NVIDIA, 2025a; Vertiv, 2026). In displays, Dai Nippon Printing doubled its OLED metal mask capacity with an 8th generation line at Kurosaki and expects its invar patents to stay protected (Dai Nippon Printing, 2024a, 2024b). DNP itself notes that some panel makers test mask free deposition, which TrendForce reports Visionox and TCL CSOT pursue, while Samsung Display and BOE keep fine metal masks on 8.6 generation lines (Dai Nippon Printing, 2024b; TrendForce, 2025).

Table 8: High performance metal applications in electronics, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | OLED fine metal masks for IT panels, at risk from mask free patterning | FeNi36 ultra thin foil |
| Grows | Current sensors and shielding in power electronics and data centres | Soft magnetic FeNi strip |
| Grows | Fab gas delivery and chamber components | High purity nickel alloy and stainless strip and tube stock |
| Stable | Glass seals, lead frames, resistance heating | FeNi42 and FeNi29Co18 strip, resistance wire |

*Sources: team assessment (Dai Nippon Printing, 2024a; Infineon Technologies, 2026; NVIDIA, 2025a; SEMI, 2026a).*

Because the filings of TSMC, Applied Materials and Lam Research name no specific alloy, suppliers reach this market through component makers. Infineon's EUR 5 billion Smart Power Fab in Dresden, opened in July 2026, shows that Europe still adds capacity in power electronics (Infineon Technologies, 2026).

Implications for suppliers:

- Partner with mask makers to qualify FeNi36 ultra thin foil rather than compete with protected processes.
- Qualify high purity alloy strip and tube with the component suppliers of fab gas systems.

## 5.7 Medical Technology

Medical technology offers the steadiest growth of all seven industries, because ageing populations rather than business cycles drive the hip and knee replacements made from cobalt chromium, titanium and zirconium alloys.

### 5.7.1 Demand, Geography and Market Size

Joint replacement is the most measurable driver of implant metal demand, as in 2023 the OECD average was 198 hip and 156 knee replacements per 100,000 people, and rates rose over the decade in nearly every OECD country (OECD, 2025). Europe's medical technology market reached about EUR 173 billion in 2025, while the United States accounts for 46.4 percent of global device sales (MedTech Europe, 2025, 2026). China is large but unprofitable for implant metals, since central purchasing cut the prices of artificial hips and knees by about 82 percent (National Healthcare Security Administration, 2022). The best measured size is at the procedure stage, about 2.75 million hip and knee replacements in 2023 in the 35 OECD members reporting both, excluding the United States and Japan, while implant alloy tonnage is not published (OECD, 2025).

### 5.7.2 Applications and Opportunities

The buyers of implant metals are few and powerful, which makes approval and reliability more important than price. Stryker reported orthopaedic revenue of USD 9,469 million in 2025, and the four leading orthopaedic groups exceed USD 20 billion of implant revenue (Johnson & Johnson, 2025; Smith & Nephew, 2026; Stryker, 2026). The mill side is volatile, as bar supplier Carpenter saw medical sales excluding surcharge fall 24 percent to USD 224.3 million in fiscal 2026 on customer destocking (Carpenter Technology, 2026).

Specifications differ by product form, as ASTM F75 covers cast CoCrMo, both investment castings such as knee femoral components and the casting alloy as ingot or bar, ASTM F1537 covers wrought CoCrMo bar, rod and wire, and ASTM F3213 covers CoCrMo made by powder bed fusion, as in the printed talus that the FDA approved in 2023 (ASTM International, 2017, 2020, 2023; U.S. Food and Drug Administration, 2023). Smith & Nephew's OXINIUM, an oxidized surface on a Zr2.5Nb alloy developed as an alternative to cobalt chromium femoral components, reaches 94.1 percent survivorship at 20 years, so it takes demand from CoCr but needs wrought Zr2.5Nb bar to ASTM F2384 from an undisclosed supplier (ASTM International, 2024, n.d.; Smith & Nephew, 2026).

Table 9: High performance metal applications in medical technology, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Cast knee femoral components | CoCrMo casting alloy, ingot and bar to ASTM F75 |
| Grows | Wrought hip and knee components | CoCrMo bar to ASTM F1537 |
| Grows | Additively manufactured implants | CoCrMo powder to ASTM F3213 |
| Grows | Oxidized zirconium femoral components | Zr2.5Nb bar to ASTM F2384 |

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Falls | Implants for China under central purchasing | CoCr bar |

*Sources: team assessment (ASTM International, 2023; National Healthcare Security Administration, 2022; Smith & Nephew, 2026).*

Implications for suppliers:

- Offer CoCrMo casting alloy to ASTM F75 and bar to ASTM F1537 to implant makers in Europe and the United States.
- Develop CoCrMo powder to ASTM F3213 and Zr2.5Nb bar to ASTM F2384, and pass the supplier audits of the device makers.

## 6. Global Market Size and Trade

Trade data show how large the market for qualified metal products is and where its value is created, before this chapter turns to trade policy.

### 6.1 Measuring the Market for Qualified Nickel Products

The only product level market that official statistics measure for Europe is the apparent consumption of nickel mill products, which runs to tens of kilotonnes. EU production plus imports minus exports came to 57.8 kt of nickel plate, sheet, strip and foil and 9.2 kt of tubes in 2024, while the bar figure rests on a confidential production cell (Eurostat, 2025a, 2025b, 2025d). Because Eurostat production values fall below the value of the contained nickel, Table 10 values these tonnes at EU border unit values, a team estimate of about EUR 1.8 to 2.8 bn for flat products and EUR 0.24 to 0.38 bn for tubes (Eurostat, 2025c). The codes cover only alloys in which nickel predominates, so the qualified market is larger, and the unit values show the conversion premium that section 6.2 explains.

Table 10: European market for nickel mill products, 2024

| Product code | EU apparent consumption | Extra EU unit value, imports and exports (EUR/kg) | Value, team estimate |
|---|---|---|---|
| Bars, rods and wire (PRODCOM 24452100) | Confidential | 25.4 and 33.4 | Not estimated |
| Plate, sheet, strip and foil (24452200) | 57.8 kt | 47.8 and 31.0 | EUR 1.8 to 2.8 bn |
| Tubes and fittings (24452300) | 9.2 kt | 41.1 and 26.1 | EUR 0.24 to 0.38 bn |

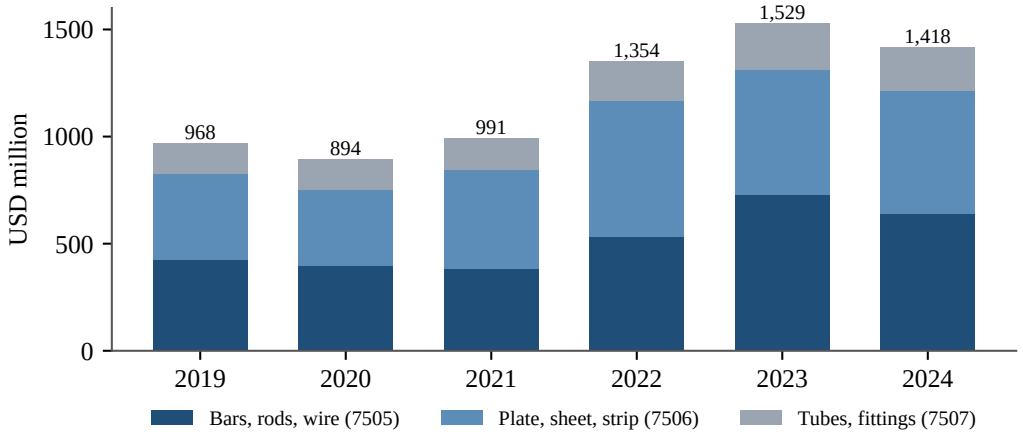
| Product code | EU apparent consumption | Extra EU unit value, imports and exports (EUR/kg) | Value, team estimate |
|---|---|---|---|
| Contained nickel, 2024 | | 15.5 | |

*Sources: Eurostat PRODCOM DS-059358, apparent consumption as sold production plus extra EU imports minus exports, team calculation of unit values and value range (Eurostat, 2025c, 2026d). Contained nickel from the World Bank price at the ECB rate (European Central Bank, 2025; World Bank, 2026a).*

## 6.2 Conversion Creates the Trade Surplus

Trade flows show that value is earned where metals are converted, and among the ten reporting countries Germany is the largest exporter of nickel plate, sheet and strip and the second largest of bars and wire, and its exports of the three mill product codes rose from USD 0.968 bn in 2019 to USD 1.418 bn in 2024, an increase of about 47 percent (United Nations Statistics Division, 2026b, 2026c, 2026d).

Figure 2: German exports of nickel mill products rose by about 47 percent from 2019 to 2024




*Source: UN Comtrade, Germany as reporter, exports to the world in current USD (United Nations Statistics Division, 2026b, 2026c, 2026d).*

Table 11: Leading exporters of nickel mill products and unwrought nickel, 2024

| HS code and product | Leading exporters among the ten reporters (USD million) | Germany net trade (USD million) |
|---|---|---|
| 7505 Nickel bars, rods, profiles, wire | United States 1,400, Germany 638, France 428, United Kingdom 402, Italy 388 (mirror: Austria 384) | +79 |

| HS code and product | Leading exporters among the ten reporters (USD million) | Germany net trade (USD million) |
|---|---|---|
| 7506 Nickel plate, sheet, strip, foil | Germany 578, United States 482, Japan 274, France 110, China 85 (mirror: Sweden 101) | +414 |
| 7507 Nickel tubes and fittings | United States 324, Germany 202, China 195, United Kingdom 168, Italy 141 | +105 |
| 7502 Unwrought nickel | China 2,068, Indonesia 862, United Kingdom 768 (mirror: Canada 1,236, Norway 1,086, Russia 1,049, Australia 981) | −736 |

*Sources: UN Comtrade reporter data for Germany, the United States, China, Japan, Korea, India, France, the United Kingdom, Italy and Indonesia (United Nations Statistics Division, 2026e). Mirror values are these ten countries' imports from a non reporter, a lower bound (United Nations Statistics Division, 2026a). Net trade is a team calculation.*

The pattern in Table 11 is the market's central economic fact. Germany imports USD 736 million more raw nickel than it exports, yet it is a net exporter of every nickel mill product, and its plate and strip sell at about twice the value of the contained metal, as do EU exports at EUR 31.0 per kg (Eurostat, 2025c; United Nations Statistics Division, 2026c; World Bank, 2026a). Reporter data alone suggest that China leads unwrought nickel exports, but mirror flows show Canada, Norway, Russia and Australia, the main class 1 origins, each shipping about USD 1 bn to the ten reporters, more than Indonesia reports (United Nations Statistics Division, 2026a). Austria and Sweden appear in the mirror data as converters on the German pattern, so value accrues where nickel is melted, rolled and drawn, which is where the corridors of section 6.3 run.

## 6.3 Corridors and Dependencies

The geography of trade mirrors the geography of the industries that buy. German nickel bar and wire go mainly to the United Kingdom and the United States, each with about USD 111 million in 2024, the engine corridors of section 5.4, while plate and strip go mainly to Austria, Italy, China and the United States, where chemical and electronics fabricators work (United Nations Statistics Division, 2026b, 2026c). Engine trade confirms the picture, with British exports of turbojets and gas turbines at USD 39.8 bn in 2024, re exports included, and trade policy now reshapes these corridors (United Nations Statistics Division, 2026e).

## 6.4 Trade Policy Reshapes Sourcing and Access

Trade policy now changes costs and market access faster than demand does, as the United States raised its Section 232 tariffs on steel and aluminium to 50 percent in 2025 and taxed the

copper content of semifinished copper products at 50 percent from 1 August 2025 (The White House, 2025). The EU Carbon Border Adjustment Mechanism, definitive since 1 January 2026, covers ferrochrome and ferronickel but not the nickel products of HS chapter 75, so it raises input costs without protecting nickel alloy mill products (Ernst & Young, 2025). The Chinese licences, Congolese cobalt quota and Indonesian ore quota of chapter 4 add supply risk, while the EU Critical Raw Materials Act lists nickel, cobalt and tungsten as strategic or critical (European Parliament and Council, 2024; Indonesia Business Post, 2026; Ministry of Commerce of the People's Republic of China, 2025a; U.S. Geological Survey, 2026e). These measures reward producers that convert close to their customers and secure several sources of each critical input, the setting in which chapter 7 projects demand to 2035.

## 7. Growth Scenarios to 2035

The final link asks how large alloy demand could become by 2035, answered by a calibrated tier 1 model and a separate tier 2 register, both team estimates (Product Strategy 2035 Team, 2026a).

### 7.1 Growth Model and Scenario Assumptions

Tier 1 writes demand as D = A × I × S, the activity driver times alloy mass per unit times the share of units using a high performance alloy, each as a ratio of 2035 to 2025 and indexed to 2025 = 100. It is calibrated to measured end use, as 11,499 t of the 20.0 kt of superalloy cobalt went to aviation in 2025, and the IEA nickel path outside energy technologies sets the envelope (Cobalt Institute, 2026; International Energy Agency, 2026b).

Table 12: Tier 1 drivers and assumptions, low to high range

| Industry | Activity driver | Growth range to 2035 | Intensity, share | Key source |
|---|---|---|---|---|
| Aerospace | New engines and aftermarket | 2.0 to 5.0% and 3.0 to 4.2% a year | Share 0.90 to 1.03 | (Airbus, 2026b) |
| Energy | Gas turbines, nuclear, other | Turbine output ratio 1.15 to 1.85 | ±1% a year | (Siemens Energy, 2025) |
| Electronics | 300 mm fab capacity, components | Index 138 in 2031, then 0 to 6% | −1.5 to +1% a year | (SEMI, 2026b) |
| Medical | Hip and knee procedures | 3.0 to 4.1% a year | CoCr share 0.90 to 1.02 | (OECD, 2025) |
| Chemical | Chemical activity | −1.0 to 2.5% a year | ±0.5% a year | (European Chemical Industry Council, 2026b) |
| Oil and gas | Upstream investment, LNG | Ratio 0.80 to 1.12 and 1.15 to 1.51 | −1.5 to +1% a year | (International Energy Agency, 2026h) |
| Automotive | Car sales, electric share | Electric share 40 to 62% in 2035 | Turbo alloy 0 to 2% a year | (International Energy Agency, 2026c) |

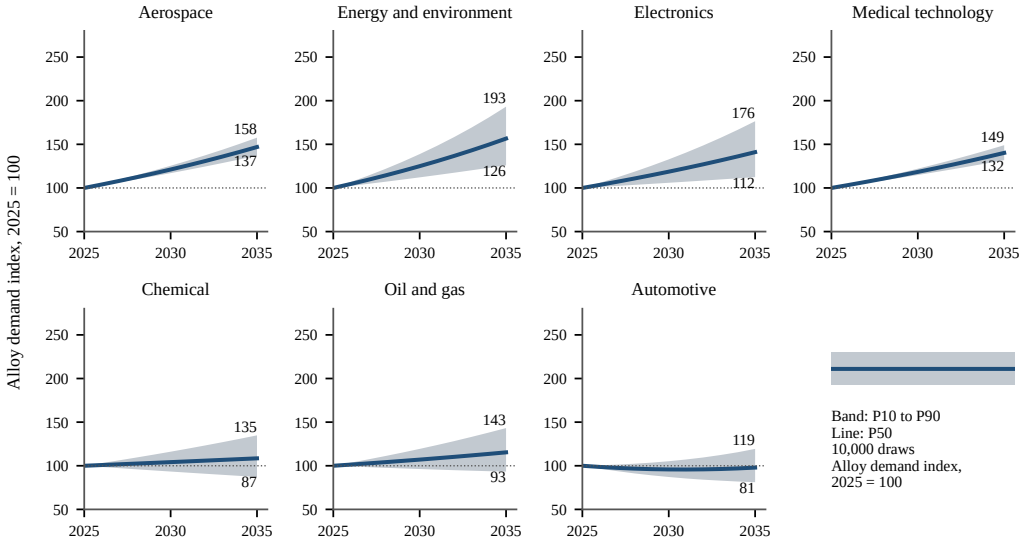
*Source: team assumptions, all parameters in Table A1.*

A backtest shows that past trends erred in both directions, because fitted to 2019 the trend underestimated gas power investment for 2020 to 2025 with a mean absolute percentage error of 36.5 percent and a bias of minus 36.0 percent, but overstated EU vehicle output by 23.0 percent (Product Strategy 2035 Team, 2026a). Half of each growth error is corrected, capped at 1 percentage point a year, and the rest widens the ranges of a Monte Carlo simulation with 10,000 draws and seed 20261108, yielding the percentiles P10, P50 and P90.

## 7.2 Results by Industry, Product, and Region

The results place growth in turbines, engines, electronics and implants, as Figure 3 and Table 13 show, while automotive stays near 98 because combustion parts fall to 66 and electric vehicle parts rise to 192 (Product Strategy 2035 Team, 2026a).

Figure 3: Energy technology and aerospace lead alloy demand to 2035, while automotive stays flat




*Source: team calculation, band P10 to P90, line P50 (Product Strategy 2035 Team, 2026a).*

Table 13: Tier 1 alloy demand index 2035 by industry (2025 = 100)

| Industry | P10 | P50 | P90 | P50 growth a year |
|---|---|---|---|---|
| Energy | 126 | 156 | 193 | 4.5% |
| Aerospace | 137 | 147 | 158 | 3.9% |
| Electronics | 112 | 141 | 176 | 3.5% |
| Medical | 132 | 140 | 149 | 3.4% |
| Oil and gas | 93 | 115 | 143 | 1.4% |
| Chemical | 87 | 109 | 135 | 0.9% |
| Automotive | 81 | 98 | 119 | −0.2% |

*Source: team calculation (Product Strategy 2035 Team, 2026a).*

Weighted into product families, the model favours precision FeNi and vacuum melted bar over flat products and tubes, although only the aerospace weight of 0.575 is measured and the others are team assumptions (Product Strategy 2035 Team, 2026a).

Table 14: Product family index 2035 (2025 = 100) and main industry weights

| Product family | Main industry weights | P10 | P50 | P90 |
|---|---|---|---|---|
| Precision FeNi strip and foil | Electronics 0.60, electric vehicles 0.30 | 134 | 157 | 185 |
| Superalloy bar, billet, remelt stock | Aerospace 0.575, energy 0.30 | 131 | 143 | 158 |
| Nickel alloy bar, rod and wire | Chemical 0.25, oil and gas 0.25 | 115 | 128 | 142 |
| Plate, sheet, strip and foil | Chemical 0.30, energy 0.20 | 113 | 125 | 139 |
| Tubes and fittings | Oil and gas 0.35, chemical 0.25 | 111 | 125 | 141 |

*Source: team calculation, weighted industry indices (Product Strategy 2035 Team, 2026a).*

In the supply overlay, IEA cobalt mine supply falls 30 percent short of 2035 demand, 36 percent if the Congo quota stays at its 2026 level, yet superalloys need only 13 to 20 percent of it, so cobalt is a cost risk rather than a volume ceiling, while the refined nickel gap of 21 percent is an upper bound and copper mines cover 62 percent of demand before scrap (Product Strategy 2035 Team, 2026a; U.S. Geological Survey, 2026e).

Regionally, engine and turbine growth sits in the United States, a fifth of superalloy cobalt, and Western Europe, while reactor tube demand follows the 40 of 81 reactors under construction in China and Asia takes more than 80 percent of chemical growth (BASF, 2026; U.S. Geological Survey, 2026e; World Nuclear Association, 2026b).

Beyond these measured markets, Table 15 registers emerging options, partly from the team research workbook, which enter tier 1 only once their signpost occurs (Product Strategy 2035 Team, 2026a, 2026b).

Table 15: Tier 2 register, not included in tier 1

| Opportunity, alloy and form | TRL | Evidence | Earliest volume | Signpost |
|---|---|---|---|---|
| Solid oxide cells, ferritic strip | 9 | Early signal | 2026 | Strip contract at Bloom Energy |
| Electrolysers, nickel strip and mesh | 8 to 9 | Early signal | 2026 | Nucera orders above EUR 356 m |
| Additive engine parts, superalloy powder | 7 to 9 | Early signal | 2027 | Part specification with volume |
| 800 VDC racks, less copper busbar | 6 to 7 | Early signal | 2027 | Eaton and Vertiv busbar launches |
| Small reactors, zirconium, Alloy 690 tube | 7 to 8 | Early signal | 2030 | Serial tube order after licence |
| Hydrogen turbines, castings and sheet | 5 to 7 | Speculative | 2030 | Order with hydrogen supply |
| Fusion magnets, Alloy C-276 tape foil | 4 to 6 | Speculative | 2030 | Magnet order for a pilot plant |

| Opportunity, alloy and form | TRL | Evidence | Earliest volume | Signpost |
|---|---|---|---|---|
| sCO2 recuperators, nickel alloy plate | 5 to 6 | Speculative | 2032 | Pilot above 10 MW |

*Sources: team register, TRL and years are team estimates (Bloom Energy, 2026; NVIDIA, 2025b; Product Strategy 2035 Team, 2026a, 2026b).*

Measured growth in four industries and these dated options define the opportunity space that chapter 8 scores.

## 8. Strategic Recommendations

The previous chapters followed the chain from material to growth, and this chapter turns their evidence into priorities for producers of high performance metals in general.

### 8.1 Priority Products, Industries, Regions, and Customers

Table 16 scores both tiers for a qualified producer that enters or expands a segment, from 1 to 5 where 5 always means attractive, weighting market 15 percent, growth 20 percent, supply security 15 percent and the other criteria 10 percent each. Table A4 derives every score, for example growth from the P50 growth bands of Table 13, and early signals qualify at most as gated options.

Table 16: Opportunity matrix, weighted scores (1 = low, 5 = high attractiveness)

| Opportunity | Ev | Mkt | Gro | Sup | Comp | Tech | Cust | Barr | Diff | Score | Priority |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Turbine remelt stock and sheet | M | 4 | 5 | 3 | 2 | 4 | 5 | 5 | 4 | 4.05 | 1 |
| Engine bar and billet | M | 5 | 4 | 3 | 2 | 5 | 5 | 4 | 4 | 4.00 | 1 |
| FeNi36 foil and FeNi strip | M | 3 | 5 | 5 | 2 | 5 | 3 | 3 | 4 | 3.90 | 1 |
| CoCr implant bar and powder | M | 3 | 4 | 2 | 3 | 4 | 4 | 5 | 3 | 3.45 | 2 |
| Sour service bar, strip, wire | M | 4 | 2 | 4 | 3 | 3 | 3 | 5 | 3 | 3.30 | 2 |
| Chemical plate, welding wire | M | 5 | 2 | 4 | 2 | 3 | 3 | 3 | 3 | 3.15 | 2 |
| Alloy 690 bar and hollows | M | 2 | 2 | 5 | 3 | 4 | 4 | 3 | 3 | 3.15 | 2 |
| Solid oxide cell strip | E | 2 | 3 | 4 | 3 | 4 | 2 | 3 | 4 | 3.10 | Option |
| Alloy C-276 fusion tape foil | E | 1 | 2 | 4 | 4 | 5 | 2 | 3 | 5 | 3.05 | Option |
| Additive engine powder | E | 2 | 3 | 3 | 2 | 4 | 4 | 3 | 3 | 2.95 | Monitor |
| Car exhaust and turbo alloys | M | 3 | 1 | 5 | 3 | 1 | 2 | 3 | 2 | 2.50 | Harvest |

*Source: team assessment, rules in Table A4 (Product Strategy 2035 Team, 2026a). Ev evidence (M measured, E early signal), Mkt market, Gro growth, Sup supply security, Comp competitive intensity (5 = low), Tech technology, Cust customers, Barr entry barriers, Diff differentiation. Priority 1 from 3.75, 2 from 3.00.*

The ranking points to vacuum melted superalloy in the United States and Europe, namely remelt stock and combustor sheet for Siemens Energy and casting houses such as CPP, and triple melt Alloy 718 and 720Li billet for the forgers of GE Aerospace and Safran. Capacity rather than metal is scarce, as Siemens Energy is sold out to 2028 and ATI quotes forging lead times beyond two years, but ATI, PCC, Carpenter, Aubert & Duval and Daido hold long term agreements, so competition scores 2 and entrants win only as qualified second sources (ATI, 2026b; Hepher et al., 2026; Siemens Energy, 2025). FeNi36 mask foil for DNP and BOE in East Asia completes priority 1 with a P50 of 157, although maskless patterning could replace invar foil (Display Daily, n.d.).

Priority 2 holds approval protected franchises, CoCr bar for Stryker and Zimmer Biomet, sour service products in the Americas, chemical plate through Asian fabricators and Alloy 690 for tube makers (Stryker, 2026; Zimmer Biomet, 2026). All three priority 1 positions keep their rank under equal weights or with entry barriers at 20 percent.

## 8.2 Recommended Actions and Implementation Roadmap

A producer should therefore let capacity follow signed demand, as Table 17 sets out with owners, peer benchmarks and gates.

Table 17: Implementation roadmap 2026 to 2035

| Initiative, owner | 2026 to 2028 | 2029 to 2035 | Benchmark and gate |
|---|---|---|---|
| 1. Superalloy, sales and quality | Qualify remelt stock and 718 billet | VIM and VAR capacity | ATI Monroe VIM 9 kt a year, Carpenter Athens capex USD 243 m. Gate: contracts for half by 2028 |
| 2. FeNi, development | Qualify foil with mask makers | Thin rolling line | Aperam Imphy VIM, EUR 160 m plan. Gate: approval by 2028 |
| 3. Franchises, sales | Keep NORSOK, F1537, ASME III | Stock in India | Alleima Mehsana SEK 220 m. Gate: Asian orders rise |
| 4. Tier 2 options, technology | Pilot lots | Scale after signpost | Carpenter powder margin 7.9%. Gate: Table 15 |
| 5. Inputs, procurement | Several cobalt sources | Closed loop revert | Daido JPY 36 bn. Gate: no source above half |

*Sources: company report benchmarks (Alleima, 2026b; Aperam, 2026; Argus Media, 2026b; Carpenter Technology, 2026; SteelOrbis, 2026).*

Focus also means dropping diesel only engine parts, European base chemical new build, electrolyser capacity before funded orders and captive nuclear cladding zirconium, and appendix A8 gives each risk an early signal and a kill criterion.

## 9. Conclusion

This study asked how supply concentration, conversion and industry needs shape the opportunities in high performance materials through 2035, and its answer is that value lies in conversion and qualification rather than in metal volume. Nickel is in surplus while cobalt, tungsten and molybdenum are bound by quotas and licences, yet the countries that convert metals into qualified bar, strip and tube earn the trade surplus, and cobalt is a cost risk rather than a volume ceiling for superalloys.

The tier 1 model then locates demand, with energy technology and aerospace at P50 indices of 156 and 147 in 2035 and automotive flat. The most attractive measured opportunities are therefore turbine remelt stock, engine billet and precision FeNi foil in the United States, Europe and East Asia, followed by approval protected franchises, while solid oxide strip, fusion magnet tape, additive powder and small reactor tubing are tier 2 options to monitor against their signposts.

These findings have limits, because official statistics stop at the mill product and most indices rest on activity drivers and team weights, not measured alloy tonnes. Within those limits the decade to 2035 rewards producers that secure their inputs, qualify early with engine and turbine makers and invest only where signed demand has arrived.

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## Appendices: Data, Sources, Calculations, and Assumptions

### A1 Definitions and the stage ladder

The report uses Mt for million tonnes, kt for thousand tonnes, t for metric tonnes, GW for gigawatts and bcm for billion cubic metres, and all prices are in current US dollars or euros of the year stated. The stage ladder separates five rungs that are never added together: mine output of contained metal, refined metal or intermediate such as nickel pig iron or cobalt hydroxide, alloy melt, mill product such as strip, sheet, plate, bar and wire, and finished component. Contained metal is kept apart from gross weight, actual output from capacity, and commodity benchmarks such as the World Bank Pink Sheet from the selling price of qualified alloy products.

### A2 Outlook model, backtest and opportunity scoring

The tier 1 model of chapter 7 is fully reproducible from the repository, and this appendix documents its formulas, parameters, backtest, simulation settings and the scoring rules of chapter 8. Every value is a team estimate unless a source is given, and running python3 model/outlook_2035.py from the repository root rewrites data/clean/model_tier1_results.csv, model_backtest.csv, model_calibration.csv, model_supply_overlay.csv, model_tier2_register.csv, research/model/outlook_results.md and Figures 3 and 4 with identical numbers, while the option --no-log skips the update of the source register (Product Strategy 2035 Team, 2026a).

#### A2.1 Formulas and simulation settings

For each industry j with components k, such as new engines and aftermarket in aerospace, the 2035 index is I(j) = 100 × Σ w(k) × A(k) × (1 + i(k))^10 × S(k) × exp(10 × (Δg(j) + σ(j) × ε)). Here A is the activity ratio 2035 to 2025, i the annual trend of alloy mass per unit, S the ratio of the high performance alloy share, w the component weight, Δg the backtest shift and σ the backtest error carried into the range, with ε a standard normal draw. The shift is Δg = −λ × e, where e is the mean annual growth error of the real backtest series of the industry, λ = 0.5 and the shift is capped at 1 percentage point a year, while σ = κ × the root mean square growth error of all series of the industry, with κ = 0.5 and a cap of 3 percentage points. A product family index is the weighted sum of the industry indices with the weights of Table A1, and cobalt tonnes multiply the 2025 base by the index and by the cobalt content ratio.

The simulation draws 10,000 values with the random seed 20261108 from triangular distributions with the low, mode and high values of Table A1. Activity drivers are linked by a Gaussian copula with a global cycle weight of 0.2 and an industry weight of 0.3, intensity and share draws are independent, and component and product weights vary around their mode in a Dirichlet distribution with concentration 40. All of these settings, λ, κ and both caps are team assumptions, because the backtest windows are short and the test period contains the pandemic.

### A2.2 Parameters and their sources

Table A1 lists every parameter of the tier 1 model with its range and evidence, and parameters without a cited source are team assumptions whose reasons are logged in research/model/outlook_results.md.

Table A1: Tier 1 model parameters, low / mode / high

| Industry | Parameter | Low / mode / high | Basis and source |
|---|---|---|---|
| Aerospace | Aftermarket growth a year | 3.0 / 3.7 / 4.2% | Sourced (Airbus, 2026b; Boeing, 2026; International Civil Aviation Organization, n.d.) |
| Aerospace | Engine output ratio 2026 to 2025 | 1.16 / 1.18 / 1.19 | Sourced, LEAP high teens growth (GE Aerospace, 2026b; Safran, 2026a) |
| Aerospace | Engine output growth 2026 to 2035 a year | 2.0 / 3.4 / 5.0% | Airbus fleet path, range team assumption (Airbus, 2026a) |
| Aerospace | Superalloy mass per engine, trend a year | −1.0 / 0.0 / 1.0% | Team assumption (Airbus, 2026a) |
| Aerospace | Superalloy share of hot section, ratio | 0.90 / 0.98 / 1.03 | Team assumption, ceramic composites and TiAl |
| Aerospace | Weights new engines and aftermarket | 0.5 and 0.5 | Team assumption (GE Aerospace, 2026a) |
| Energy | Gas turbine output ratio | 1.15 / 1.58 / 1.85 | Siemens Energy path 17 to 30 GW (International Energy Agency, 2026j; Siemens Energy, 2025, 2026) |

| Industry | Parameter | Low / mode / high | Basis and source |
|---|---|---|---|
| Energy | Nuclear fleet and tube ratio | 1.05 / 1.20 / 1.45 | Derived (International Atomic Energy Agency, 2026; Westinghouse Electric, 2026; World Nuclear Association, 2026a) |
| Energy | Other equipment growth a year | 2.0 / 3.0 / 3.6% | Electricity demand proxy, team assumption |
| Energy | Alloy mass per GW, trend a year | −1.0 / 0.0 / 1.0% | Team assumption |
| Energy | Alloy share of hardware, ratio | 0.97 / 1.00 / 1.03 | Team assumption |
| Energy | Weights turbines, nuclear, other | 0.55, 0.25, 0.20 | Team assumption |
| Electronics | Realisation of SEMI path to 2031 | 0.85 / 1.00 / 1.05 | SEMI index 138 in 2031 (SEMI, 2026b) |
| Electronics | Capacity growth 2031 to 2035 a year | 0.0 / 3.0 / 6.0% | Team assumption (SEMI, 2026b) |
| Electronics | Components and masks growth a year | 0.0 / 2.5 / 6.0% | Team assumption (Dai Nippon Printing, 2024b; ZVEI, 2026) |
| Electronics | Alloy per unit, trend a year | −1.5 / 0.0 / 1.0% | Team assumption (NVIDIA, 2025a) |
| Electronics | Alloy share, ratio | 0.95 / 1.00 / 1.05 | Team assumption (Dai Nippon Printing, 2024b) |
| Electronics | Weights equipment and components | 0.6 and 0.4 | Team assumption (SEMI, 2026a) |
| Medical | Hip and knee procedures growth a year | 3.0 / 3.5 / 4.1% | Derived, 30 OECD countries 2013 to 2024 (OECD, 2025) |
| Medical | Other medical uses growth a year | 1.8 / 3.0 / 5.6% | MedTech Europe value growth (MedTech Europe, 2025, 2026) |
| Medical | Alloy per procedure, trend a year | −0.5 / 0.0 / 0.5% | Team assumption |

| Industry | Parameter | Low / mode / high | Basis and source |
|---|---|---|---|
| Medical | CoCr and nickel share, ratio | 0.90 / 0.97 / 1.02 | Team assumption, oxidised zirconium and titanium |
| Medical | Weights implants and other | 0.8 and 0.2 | Team assumption (Stryker, 2026) |
| Chemical | Activity growth a year | −1.0 / 1.0 / 2.5% | Team assumption (BASF, 2026; European Chemical Industry Council, 2026a, 2026b) |
| Chemical | Alloy per output, trend a year | −0.5 / 0.0 / 0.5% | Team assumption |
| Chemical | Alloy share, ratio | 0.97 / 1.00 / 1.03 | Team assumption |
| Oil and gas | Real upstream investment ratio | 0.80 / 1.00 / 1.12 | IEA scenarios (International Energy Agency, 2025c, 2026h) |
| Oil and gas | LNG trade ratio | 1.15 / 1.35 / 1.51 | IEA, 560 to 880 bcm (International Energy Agency, 2025c, 2026f) |
| Oil and gas | Alloy per unit, trend a year | −1.5 / −0.5 / 1.0% | Team assumption, clad pipe saves material |
| Oil and gas | Corrosion resistant alloy share, ratio | 0.97 / 1.00 / 1.03 | Team assumption |
| Oil and gas | Weights upstream and LNG | 0.7 and 0.3 | Team assumption (International Energy Agency, 2026i) |
| Automotive | New car sales ratio | 0.95 / 1.00 / 1.10 | Team assumption (European Automobile Manufacturers' Association, 2026a; International Energy Agency, 2025d) |
| Automotive | Electric share of sales 2035 | 40 / 50 / 62% | IEA scenarios (International Energy Agency, 2025d, 2026c) |
| Automotive | Heat resistant alloy per combustion car, trend a year | 0.0 / 1.0 / 2.0% | Team assumption (Garrett Motion, 2026) |

| Industry | Parameter | Low / mode / high | Basis and source |
|---|---|---|---|
| Automotive | Nickel alloy share of turbo parts, ratio | 0.95 / 1.00 / 1.03 | Team assumption |
| Automotive | Precision alloy per electric car, trend a year | −1.0 / 0.0 / 2.0% | Team assumption |
| Automotive | FeNi share of motor and sensor parts, ratio | 0.90 / 1.00 / 1.10 | Team assumption |
| Automotive | Weights thermal and electric vehicle parts | 0.75 and 0.25 | Team assumption (Garrett Motion, 2026) |
| All | Nickel content of alloy class, ratio | 0.98 / 1.00 / 1.02 | Team assumption |
| All | Cobalt content, aerospace and other, ratio | 0.92 / 1.00 / 1.04 and 0.90 / 1.00 / 1.03 | Team assumption (CMOC Group, 2026) |
| Families | Superalloy bar, billet and remelt stock | Aerospace 0.575, energy 0.300, car thermal 0.050, oil and gas 0.040, chemical 0.035 | Aerospace weight measured (Cobalt Institute, 2026) |
| Families | Nickel alloy bar, rod and wire | Chemical 0.25, oil and gas 0.25, energy 0.20, aerospace 0.15, medical, car thermal, electronics 0.05 each | Team assumption (Alleima, 2026a) |
| Families | Plate, sheet, strip and foil | Chemical 0.30, energy 0.20, aerospace 0.20, oil and gas 0.15, car thermal 0.10, electronics 0.05 | Team assumption (ATI, 2026a) |
| Families | Tubes and fittings | Oil and gas 0.35, chemical 0.25, energy 0.25, aerospace, car thermal, medical 0.05 each | Team assumption (Alleima, 2026a; International Atomic Energy Agency, 2026) |
| Families | Precision FeNi strip and foil | Electronics 0.60, electric vehicle parts 0.30, medical 0.10 | Team assumption (Dai Nippon Printing, 2024b; International Energy Agency, 2026c) |

*Sources: model/outlook_2035.py and research/model/outlook_results.md (Product Strategy 2035 Team, 2026a). The calibration in data/clean/model_calibration.csv reproduces the 20.0 kt of superalloy cobalt in 2025 exactly and leaves the IEA nickel and cobalt totals for 2035 with plausible remainders (International Energy Agency, 2026b).*

### A2.3 Backtest

The backtest fits a log linear trend to each series up to 2019 and compares its projection for 2020 to the latest year with the outcome, as Table A2 shows. Only real series correct the growth rate, while nominal OECD STAN output series only widen the ranges, because price inflation after 2021 would otherwise appear as volume.

Table A2: Backtest of tier 1 drivers, trend to 2019 against outcome

| Series | Industry | Basis | Test years | MAPE % | Bias % | Growth error, pp a year | Corrects growth |
|---|---|---|---|---|---|---|---|
| IEA upstream investment | Oil and gas | Real | 2020 to 2025 | 17.3 | −7.6 | −3.45 | Yes |
| IEA gas power investment | Energy | Real | 2020 to 2025 | 36.5 | −36.0 | −20.68 | Yes |
| IEA nuclear investment | Energy | Real | 2020 to 2025 | 50.3 | −50.3 | −16.42 | Yes |
| OECD hip and knee procedures | Medical | Counts | 2020 to 2024 | 9.7 | 9.2 | −0.53 | Yes |
| OECD STAN C30 output | Aerospace | Nominal | 2020 to 2023 | 14.9 | 14.9 | 0.31 | No |
| OECD STAN C26 output | Electronics | Nominal | 2020 to 2023 | 7.5 | −6.2 | −2.58 | No |
| OECD STAN C27 output | Electronics | Nominal | 2020 to 2023 | 12.8 | −11.0 | −5.65 | No |
| OECD STAN C20 output | Chemical | Nominal | 2020 to 2023 | 13.8 | −9.9 | −4.31 | No |
| OECD STAN C29 output | Automotive | Nominal | 2020 to 2023 | 14.8 | 14.8 | −1.07 | No |
| Eurostat C29 volume, EU27 | Automotive | Real | 2020 to 2025 | 23.0 | 23.0 | 3.22 | Yes |
| Eurostat C29 volume, DE, ES, FR, IT | Automotive | Real | 2020 to 2025 | 17.2 to 49.1 | 17.2 to 49.1 | 2.02 to 7.02 | No |
| IEA WEI 2025 against WEI 2026 | Oil and gas | Real | 2025 | 5.2 | 5.2 | n/a | No |

*Sources: team calculation (Product Strategy 2035 Team, 2026a) on IEA, OECD and Eurostat data (Eurostat, 2026b; International Energy Agency, 2026j; OECD, 2025, 2026). Bias is the mean of projection minus outcome over outcome. The growth error is the trend slope minus the realised slope.*

The resulting corrections are a shift of plus 1.0 percentage point a year for energy and for oil and gas, plus 0.27 points for medical technology and minus 1.0 point for automotive, with no

shift for aerospace, chemicals and electronics. Without the shift, the P50 indices would be 142 for energy, 104 for oil and gas, 137 for medical technology and 108 for automotive, against the earlier fixed rate workbook values of 148, 105, 128 and 86, which the model replaces (Product Strategy 2035 Team, 2026a).

### A2.4 Further tier 2 candidates

Table 15 in chapter 7 shows eight options, and Table A3 adds the remaining register rows and the team research workbook candidates that the register has not yet assessed, none of which enters tier 1.

Table A3: Further tier 2 candidates

| Opportunity | Alloy and product form | TRL and evidence | Earliest volume | Signpost or status |
|---|---|---|---|---|
| Molten salt reactor vessels and piping | NiMoCr plate and tube | 3 to 5, speculative | 2035 or later | Construction permit and ASME code case |
| Superhot and enhanced geothermal casing | Nickel alloy tube and clad pipe | 4 to 6, speculative | 2032 to 2035 | Well above 400 °C with a published casing alloy |
| Liquid hydrogen aircraft tanks | Aluminium sheet, austenitic tube | 3 to 4, speculative | 2038 or later | Airframer programme launch |
| Concentrating solar receivers, Gen 3 | Nickel alloy plate, tube, forgings | Not assessed | Not assessed | Workbook candidate |
| Fuel cell bipolar plates, stationary PEM | Coated stainless thin strip | Not assessed | Not assessed | Workbook candidate |
| Robotic surgery components | NiTi wire and precision tube | Not assessed | Not assessed | Workbook candidate |
| Smaller cardiovascular delivery systems | CoNiCrMo fine wire and tube | Not assessed | Not assessed | Workbook candidate |
| Compact electronics cooling | CuCrZr and copper powder | Not assessed | Not assessed | Workbook candidate |

*Sources: team register and team research workbook, TRL and years are team estimates (Product Strategy 2035 Team, 2026a, 2026b).*

### A2.5 Scoring rules for the opportunity matrix

The matrix in Table 16 takes the perspective of a qualified producer that already melts and rolls high performance alloys and enters or expands a segment, and Table A4 states how each score from 1 to 5 is derived. Scores of 2 and 4 lie between the anchors shown, early signal rows score growth on their earliest volume year because the model does not quantify them, and the weighted score is the sum of each score times its weight.

Table A4: Scoring rules, weights and evidence

| Criterion, weight | Evidence used | Score 5 | Score 3 | Score 1 |
|---|---|---|---|---|
| Market attractiveness, 15% | Measured 2025 pool and unit value (chapters 5 and 6) | Largest measured pools, such as aviation cobalt or EU flat products of EUR 1.8 to 2.8 bn | Measured niche with premium unit value | No measured volume yet |
| Growth to 2035, 20% | P50 growth a year of the most specific index in Tables 13 and 14 or driver in Table A1, tier 2 by earliest volume | 4.0% a year or more (4 from 3.0%) | 2.0 to 2.9% a year (2 from 0%), or tier 2 volume by 2028 | Below 0%, or tier 2 volume after 2033 |
| Supply security, 15% | Concentration, quotas and licences of the main inputs (chapter 4, supply overlay) | Nickel, iron or chromium inputs without export restriction | Cobalt as a minor element under the Congo quota | An input under an active licence halt |
| Competitive intensity, 10% | Number of qualified suppliers and long term agreements (A6) | Demand above qualified capacity, no dominant incumbent | Several qualified suppliers competing on service | Price competition with new Asian capacity (2 for an oligopoly with agreements) |
| Technology attractiveness, 10% | Process route and how many producers master it | Triple melt, foil below 0.1 mm or similar rare routes | Standard grade on a qualified route | Commodity grade in decline |
| Customer attractiveness, 10% | Buyer backlog, concentration and pricing (chapter 5) | Named buyers with sold out capacity or growing backlog | Fragmented or cyclical buyers | Collapsing orders or state price cuts |
| Entry barriers, 10% | Time to approval and protection afterwards (A7) | Passable within about two years and protective thereafter (4 for three to five years) | Programme specific or moderate approvals | Captive supply chain closed to outsiders |
| Differentiation potential, 10% | Scope for grade, tolerance or service advantage | Customer specific grades or tolerances | Partly standardised products | Interchangeable products |

*Source: team scoring rules (Product Strategy 2035 Team, 2026a). Priority 1 from a weighted score of 3.75, priority 2 from 3.00, and early signal rows are at most gated options. Under equal weights the three priority 1 rows score 4.00, 4.00 and 3.75, and with entry barriers at 20 percent they score 4.14, 4.00 and 3.82.*

## A3 Data sources, product codes and reproducibility

All structured data were pulled with the project's fetch script, which stores raw files, writes clean tables and logs every query with its date and URL in the source register of more than 1,000 entries. UN Comtrade data cover the product codes HS 7502, 7504, 7505, 7506, 7507, 8105, 8109, 7219, 7220, 7223, 7409, 7606, 8311 and 8411 for ten reporting countries from 2019 to 2024, and Eurostat PRODCOM data cover fifteen nickel, cobalt, stainless, copper and aluminium product codes for the EU and five producer countries. USGS Mineral Commodity Summaries 2026, the World Bank Pink Sheet, OECD STAN and ICIO, UNIDO INDSTAT for Germany and the filings of 61 companies complete the evidence base.

The value range of Table 10 in chapter 6 is a team estimate that multiplies EU apparent consumption in tonnes by the lower and the higher of the extra EU export and import unit values of 2024, because the Eurostat production values fall below the value of the contained nickel. Flat products give 57.8 kt × EUR 31.0 per kg = EUR 1.79 bn and 57.8 kt × EUR 47.8 per kg = EUR 2.76 bn, and tubes give 9.2 kt × EUR 26.1 per kg = EUR 0.24 bn and 9.2 kt × EUR 41.1 per kg = EUR 0.38 bn, while bars are not valued because their production cell is confidential (Eurostat, 2025c, 2026d). The unit values are trade value divided by trade quantity in PRODCOM DS-059358, so the range reflects the mix of grades traded across the EU border rather than a list price.

## A4 Limitations and further research

The study has three limitations that further research on this market should address. Official statistics stop at the mill product and mix pure nickel with nickel alloys, so the market for specific alloys can only be approximated, and a producer level survey or customs data at the eight digit level would sharpen these estimates. The tier 1 model rests on published activity drivers, but its intensity, share and product family weights are team assumptions, nickel alloy tonnes by industry are not published, and the backtest windows are short, so the model should be rerun when the IEA World Energy Outlook 2026, the next Airbus and Boeing outlooks and new SEMI forecasts appear. Finally, the trade analysis covers ten large reporting countries, so a world mirror data analysis that includes Russia, Sweden and Austria would complete the picture of global corridors.

## A5 Sector market indicators

The following tables collect the market indicators behind the industry analysis in chapter 5, with their sources and observation years.

Table A5: Automotive market indicators, 2025 and 2035

| Indicator | 2025 | 2035 | Source basis |
|---|---|---|---|
| World motor vehicle production | 96.4 million | No official forecast | OICA |
| Electric share of new car sales, world | 25% | About 50% | IEA, both exploratory scenarios |
| Electric share of car sales, China | Close to 55% | Above 90% | IEA |
| Electric share of car registrations, EU | 26.8% (BEV 17.4% plus PHEV 9.4%, team sum) | Above 90% of light duty sales | ACEA, IEA Stated Policies |

| Indicator | 2025 | 2035 | Source basis |
|---|---|---|---|
| Turbocharger production | Close to 50 million units | Back toward 2022 level by 2030 | Garrett citing S&P Global Mobility |

*Sources: OICA (2026), ACEA (2026), IEA Global EV Outlook 2026, Garrett Motion 10-K 2025 (European Automobile Manufacturers' Association, 2026a, 2026b; Garrett Motion, 2026; International Energy Agency, 2026c; International Organization of Motor Vehicle Manufacturers, 2026).*

Table A6: Oil and gas indicators, today and 2035

| Indicator | Today | 2035 | Source basis |
|---|---|---|---|
| Upstream oil and gas investment | USD 570 bn (2025), USD 546 bn (2026) | Near today's level (Stated Policies), about 10% higher (Current Policies) | IEA |
| Oil demand | 100 mb/d (2024) | Peak near 102 mb/d around 2030, back near 100 mb/d | IEA Stated Policies |
| LNG trade | 560 bcm (2024) | 880 bcm (Current Policies) | IEA |
| Upstream growth regions 2026 | Africa +12%, Central and South America +10% | Continued shift | IEA |

*Sources: IEA World Energy Outlook 2025 and World Energy Investment 2026 (International Energy Agency, 2025c, 2026h).*

Table A7: Energy technology indicators relevant to high performance metals

| Indicator | Latest value | Direction to 2035 | Metal and product form |
|---|---|---|---|
| Siemens Energy gas turbine capacity | About 17 GW (FY2024), sold out to FY2028 | Above 30 GW a year from FY2028 | Superalloy bar, billet and forging stock |
| Reactors under construction | 81 reactors, 88.5 GWe gross, 40 in China | 710 GW installed in 2035 (IEA Net Zero) | Zirconium cladding tube, Alloy 690 tube |
| Installed electrolysis | More than 4 GW (2025) | Committed 2030 output 4.3 Mt of hydrogen | Nickel strip and mesh |

| Indicator | Latest value | Direction to 2035 | Metal and product form |
|---|---|---|---|
| Solid oxide fuel cells, Bloom Energy | 1 GW a year, doubling to 2 GW by end 2026 | Driven by data centre power | Interconnect material not disclosed |

*Sources: Siemens Energy capital markets day 2025, World Nuclear Association (2026), IEA World Energy Outlook 2025 and Global Hydrogen Review 2026, Bloom Energy results (Bloom Energy, 2026; International Energy Agency, 2025c, 2026e; Siemens Energy, 2025; World Nuclear Association, 2026b).*

Table A8: Aerospace market indicators, today and 2035

| Indicator | Today | 2035 | Source basis |
|---|---|---|---|
| Passenger fleet, aircraft of 100 seats or more | 23,310 (2025) | About 32,600 (team interpolation) | Airbus GMF 2026 |
| Commercial fleet, wider definition | About 28,000 (2025) | About 38,000 | Boeing CMO 2026 |
| New generation share of fleet | 39% (2025) | 82% | Airbus |
| Aviation services market | USD 209 bn (2026) | USD 422 bn in 2045, 3.8% a year | Airbus GSF 2026 |
| Cobalt used in aviation | 11,499 t (2025) | Grows with fleet and engine output | Cobalt Institute |
| World military expenditure | USD 2,887 bn (2025), Europe +14% | Rising in Europe | SIPRI |

*Sources: Airbus (2026), Boeing (2026), Cobalt Institute (2026), SIPRI (2026) (Airbus, 2026a, 2026b; Boeing, 2026; Cobalt Institute, 2026; Stockholm International Peace Research Institute, 2026). The 2035 passenger fleet is a team estimate on a constant growth path and not an Airbus figure.*

Table A9: Chemical industry indicators

| Indicator | Value | Implication for alloy demand |
|---|---|---|
| EU27 chemical sales, 2024 | About EUR 635 bn | Large installed base needing maintenance |
| EU share of world chemical sales | 13% (2024), from 21% (2009) | New build moves out of Europe |

| Indicator | Value | Implication for alloy demand |
|---|---|---|
| EU chemical production index, 2025 | Down 3.2% | Fewer debottlenecking projects |
| EU capacity utilisation, Q3 2026 | 75.0% against 81.3% average | Closures, not expansions |
| German utilisation, 2025 | 72.5% | Weakest home market |
| Asia Pacific share of growth to 2035 | Over 80% (BASF) | Export projects through fabricators |

*Sources: Cefic (2025, 2026), Eurostat, VCI, BASF Report 2025 (BASF, 2026; European Chemical Industry Council, 2025a, 2025b, 2026b; Eurostat, 2026c; Verband der Chemischen Industrie, 2026).*

Table A10: Electronics indicators relevant to high performance metals

| Indicator | Value | Relevance |
|---|---|---|
| World semiconductor sales | USD 795.6 bn (2025), USD 1.51 tn (2026 forecast) | Activity, not metal volume |
| 300 mm fab equipment spending | USD 149 bn (2026), USD 227 bn (2029) | Gas delivery and chamber parts in high purity alloys |
| Copper busbar in a 1 MW rack at 54 VDC | Up to 200 kg | Copper and copper alloy strip and bar |
| German electrical and electronics value added | EUR 112.2 bn (2023), real output flat in 2025 | Home market for sensor and seal alloys |
| DNP metal mask capacity | Doubled by the 8th generation Kurosaki line | FeNi36 (invar) ultra thin foil |

*Sources: WSTS (2026), SEMI (2026), NVIDIA (2025), OECD STAN and ZVEI, Dai Nippon Printing (Dai Nippon Printing, 2024a; NVIDIA, 2025a; OECD, 2026; SEMI, 2026a; World Semiconductor Trade Statistics, 2026; ZVEI, 2026).*

Table A11: Medical technology indicators

| Indicator | Value | Relevance |
|---|---|---|
| Hip replacements per 100,000 people, OECD average | 198 (2023) | Volume driver for CoCr and zirconium bar |
| Knee replacements per 100,000 people, OECD average | 156 (2023) | Volume driver for CoCr bar and powder |

| Indicator | Value | Relevance |
|---|---|---|
| European medtech market | About EUR 173 bn (2025) | Devices and diagnostics, not metal |
| Stryker orthopaedics revenue | USD 9,469 m (2025) | Largest implant buyer |
| Zimmer Biomet knees and hips | USD 3,322 m and USD 2,094 m (2025) | Names cobalt chrome as a purchased commodity |
| China volume based procurement | Implant prices cut by about 82% (2022) | Volume without margin |

*Sources: OECD Health at a Glance 2025, MedTech Europe, company filings, National Healthcare Security Administration (MedTech Europe, 2026; National Healthcare Security Administration, 2022; OECD, 2025; Stryker, 2026; Zimmer Biomet, 2026).*

### A6 Peer benchmark of specialty metal producers

The peer benchmark supports the competitive assessment in section 8.1 and shows the margins and current capacity moves of leading specialty metal producers.

Table A12: Peer benchmark of specialty metal producers

| Company | Revenue, latest year | Margin | Capacity move |
|---|---|---|---|
| Howmet Aerospace | USD 8,252 m (2025) | 24.8% operating | Engine products capacity, USD 453 m capex |
| Carpenter Technology | USD 3,124 m (FY2026) | 22.5% operating | Athens, Alabama expansion, capex USD 355 to 375 m guided |
| ATI | USD 4,587 m (2025) | 14.0% operating | Fifth VIM furnace in Monroe, about 9 kt a year from 2027 |
| PCC including Special Metals | USD 10.8 bn (2025) | Not published | Aerospace revenue up 7.5% |
| Aubert & Duval | EUR 844 m (2024) | Not published | Owned by Airbus, Safran and Tikehau |
| Alleima | SEK 18,630 m (2025) | 8.3% adjusted EBIT | Fourth VAR furnace at Sandviken |
| Outokumpu | EUR 5,468 m (2025) | 3.1% adjusted EBITDA | About 13 kt high nickel flat at Avesta, about EUR 150 m |
| Aperam | EUR 6,080 m (2025) | 5.6% adjusted EBITDA | New VIM complex at Imphy within EUR 160 m programme |

| Company | Revenue, latest year | Margin | Capacity move |
|---|---|---|---|
| Daido Steel | JPY 578 bn (FY2025) | 7.3% operating | JPY 36 bn high alloy melting and forging base |
| Nippon Yakin Kogyo | JPY 151 bn (FY2025) | 7.3% operating | Sales down 12% on weak high nickel demand |

*Sources: company filings and reports (Alleima, 2026a; Aperam, 2026; Argus Media, 2026b; Armees.com, 2025; ATI, 2026a; Berkshire Hathaway, 2026; Carpenter Technology, 2026; Daido Steel, 2026; Howmet Aerospace, 2026; Nippon Yakin Kogyo, 2026; Outokumpu, 2026a, 2026b). Margins as reported or calculated from reported figures, fiscal years differ.*

### A7 Qualification gates by industry

The qualification gates below determine how quickly a producer can enter each application, and they underlie the entry barrier scores of Table 16 and the gates of Table 17.

Table A13: Qualification gates by industry

| Industry | Approval gate | Who approves | Implication |
|---|---|---|---|
| Aerospace | AMS specifications, NADCAP accredited processes, part level approval | Engine makers and forgers | Years per part family, start in 2026 |
| Energy | Casting house and turbine maker approvals, ASME III and RCC-M for nuclear | Turbine makers, utilities, tube makers | Nuclear approvals only with a named programme |
| Oil and gas | NORSOK M-650, API 6ACRA and API 5LD, operator approval lists | Operators, pipe and equipment makers | Approval lists are the moat |
| Chemical | Pressure equipment rules (VdTÜV, AD 2000, ASME), corrosion test data | Fabricators, plant operators | Life cycle cost evidence wins |
| Medical | ASTM F75 and F1537 material standards, supplier audits | Implant makers with ISO 13485 | Few buyers, long relationships |

*Source: team summary of the standards named in chapter 5.*

### A8 Risks and kill criteria

The roadmap in Table 17 carries known risks, and each of them can be met with a defined response and a point at which a producer should stop, as Table A14 shows.

Table A14: Main risks, responses and kill criteria

| Risk | Early signal | Response | Kill criterion |
|---|---|---|---|
| Qualification slower than planned | Part approvals behind schedule | Focus on repair and casting stock, which qualify faster | No long term agreement by end 2028, so no capacity step |
| Substitution by ceramic matrix composites and titanium aluminide | New engine variants with lower nickel content | Join customer development programmes early | Falling hot section nickel content in new programmes |
| New Asian superalloy capacity | Japanese and Chinese producers win Western approvals | Speed and service close to European customers | Prices below conversion cost in tenders |
| Cobalt, tungsten and molybdenum supply | Quota and licence changes | Several sources, revert recovery, stocks | Single source above half of an input |
| European demand weakness | Chemical utilisation below 75%, falling engine volumes | Shift sales to Asia and the United States | Contribution margin below plan for two years |

*Source: team assessment based on chapters 4 to 7 (European Chemical Industry Council, 2026b; SteelOrbis, 2026).*